ClipSAT : ACT International Subject Test: Mathematics 2 Level 2

Unit 5.01: Trigonometry, Identities & Polar Coordinates

Name: _____

Q1. A point has polar coordinates r, theta = left6, frac{2pi}{3}right. What are the Cartesian coordinates x, y of this point?

[ ] A: -3, 3sqrt{3}
[ ] B: -3sqrt{3}, 3
[ ] C: 3, -3sqrt{3}
[ ] D: 3sqrt{3}, -3

Q2. In triangle ABC, the length of side AB is x, the length of side BC is x + 3, and the length of side AC is 7. If angle B = 60^circ, what is the value of x?

[ ] A: 5
[ ] B: 8
[ ] C: 4
[ ] D: 6

Q3. In the polar coordinate plane, points A and B lie on the curve r = 4sin2theta such that their polar angles satisfy 0 < theta_A < theta_B < frac{pi}{2} and theta_B - theta_A = frac{pi}{3} . What is the maximum possible area of triangle OAB , where O is the origin?

[ ] A: sqrt{3}
[ ] B: 2
[ ] C: 2sqrt{3}
[ ] D: 4sqrt{3}

Q4. A point in the polar coordinate system has coordinates r, theta = left6, frac{5pi}{6}right. Which of the following represents the Cartesian coordinates x, y of this point?

[ ] A: -3sqrt{3}, 3
[ ] B: -3, 3sqrt{3}
[ ] C: 3sqrt{3}, -3
[ ] D: -3sqrt{3}, -3

Q5. In triangle ABC, the lengths of sides AB and BC are 5 and 8, respectively, and the measure of angle B is 60^circ. What is the length of the altitude from vertex B to side AC?

[ ] A: frac{20sqrt{3}}{7}
[ ] B: frac{40sqrt{3}}{7}
[ ] C: frac{10sqrt{3}}{7}
[ ] D: frac{20}{7}

Q6. In the polar coordinate system, point A has polar coordinates r_A, theta_A lying on the curve r = 6sintheta, and point B has polar coordinates r_B, theta_B lying on the curve r = 8costheta, with r_A > 0 and r_B > 0. If theta_A - theta_B = frac{pi}{6}, what is the maximum possible area of triangle OAB, where O is the pole origin?

[ ] A: 6
[ ] B: 6sqrt{3}
[ ] C: 9
[ ] D: 12

ClipSAT : ACT International Subject Test: Mathematics 2 Level 2

Unit 5.01: Trigonometry, Identities & Polar Coordinates

Name: _____

Q7. A point in the polar coordinate plane has coordinates r, theta = left4, frac{5pi}{6}right . Which of the following gives the rectangular coordinates x, y of this point?

[ ] A: -2sqrt{3}, 2

[ ] B: -2, 2sqrt{3}

[ ] C: 2, -2sqrt{3}

[ ] D: 2sqrt{3}, -2

Q8. In triangle ABC , the side lengths are AB = 5 and BC = 4 , and angle B is an acute angle. If cos2B = -frac{31}{32} , what is the length of side AC ?

[ ] A: 6

[ ] B: sqrt{46}

[ ] C: sqrt{41}

[ ] D: sqrt{31}